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A Cross-Layer SDN Orchestration Architecture for Stateful Edge Services: Design and Experimental Characterization

Pedro Ojo Oriakhi

Master's in Telecommunications and Computer Engineering

Supervisors:

Ph.D. Professor José Moura, Assistant Professor,
Iscte - Instituto Universitário de Lisboa

Ph.D. Professor Pedro Ramos, Assistant Professor,
Iscte - Instituto Universitário de Lisboa

Month, Year

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TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

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Acknowledgments

Write here your acknowledgements and, if applicable, any grants or sources of funding.

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Resumo

Palavras-Chave:

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Abstract

Write here the abstract in English. There is a 250 word limit.

Keywords: *Keywords; separated; by; semicolons*

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CHAPTER 1

Introduction

Orchestrating edge web services means dealing with coordination delays between routing, auto-scaling and monitoring components. This has been shown to negatively impact the overall infrastructure and even QoS due to the unaccounted effects each module has on each other. In this chapter the context, motivation and problem statement are explained in 1.1 with the objectives presented in 1.2. The research questions that emerged are available in 1.3. The methodology used to develop the thesis is detailed in 1.4, the main contributions in 1.5 and finally the outlining of the rest of the 7 chapters of the thesis are presented in 1.6.

1.1. Context, Motivation, and Problem Statement

In the year 2025 global internet traffic surprassed 7 exabytes (1 exabyte = 1 000 000 000 gigabytes), this corresponds to a more than double increase in the amount of internet traffic generated since 2020 across fixed and mobile networks, this is represented in Figure 1.1. [1]. This increase has been driven by the technological discoveries and developments such as streaming, social media, IoT and an ever growing number of connected devices.

These platforms and services are very often built and or deployed on the cloud, where resources are abundant.

1.2. Objectives

1.3. Research Questions

1.4. Research Methodology

1.5. Main Contributions of this Dissertation

1.6. Dissertation Structure

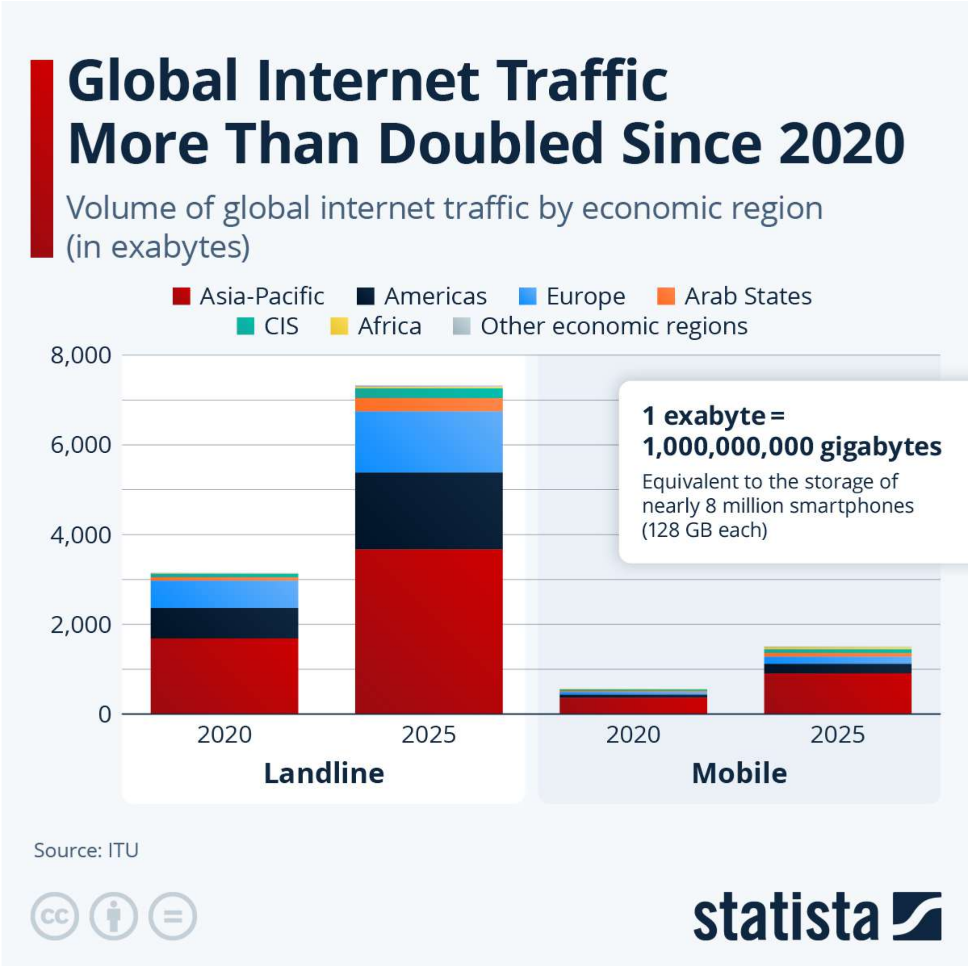

Figure 1.1. Global internet traffic between 2020 and 2025.

CHAPTER 2

Background and Related Work

2.1. Review Methodology

2.2. The Unexamined Default: Three-Layer Separation

2.3. Auto Scaling

2.4. Load Balancing on SDN

2.5. Monitoring & Telemetry

2.6. Resource Orchestration on SDN

2.7. Summary and Research Gaps

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CHAPTER 3

Architecture and Design of Proposed Solution for Edge Services

This chapter presents the design of the proposed cross layer sdn orchestration arquitecture for edge services. It stems from the research gaps explored in the 2 and brings to the fold the need for an architecture that allows for each mentioned component to be independently managed and tested.

3.1. Design Requirements

3.2. Architecture

3.3. Distributed Elastic Allocation of Resources

3.3.1. Services

3.3.2. Data

3.4. Monitoring and Decision Engine

3.5. Control Workflow

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CHAPTER 4

Implementation of Proposed Solution

4.1. Experimental Infrastructure

4.2. Distributed Elastic Allocation of Resources

4.2.1. Services

4.2.2. Data

4.3. Monitoring and Decision Engine

4.4. Control Workflow

4.5. Implementation Validation

CHAPTER 5

Evaluation Methodology

5.1. Experimental Objectives

5.2. Evaluation Scenarios

5.3. Performance Metrics

5.4. Experimental Procedure with Results Statistical Analysis

CHAPTER 6

Experimental Results

CHAPTER 7

Conclusions and Future Work

7.1. Conclusions

7.2. Research Contributions

7.3. Limitations

7.4. Future Research Directions

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References

[1] International Telecommunication Union, "Facts and figures 2025: Internet traffic," ITU, Oct. 2025. Accessed: Jul. 13, 2026. [Online]. Available: https://www.itu.int/itu-d/reports/statistics/2025/10/15/ff25-internet-traffic/